PEARL AYEM

Software & Climate Risk Lead

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PROFILE

Working at the intersection of climate science, software, and risk, I write production code for the climate, nature and supply-chain models that FTSE100 boards use to plan their next decade, lead applied AI work in agricultural risk, and have helped design data platforms launched at COP28 by heads of state. Most of my career has been spent moving between technical, scientific, policy and commercial worlds, and it's the kind of work I love most: deeply technical, genuinely useful, silently impactful.

CORE SKILLS

Software Engineering: Python, SQL, C++; Git, CI/CD, Docker, REST APIs, testing

Cloud & Data: Google Cloud (BigQuery), AWS (EC2, S3); Power BI, Tableau

Applied AI: LangSmith, LangGraph, agentic workflows, Gemini (Vertex AI, NotebookLM); CNNs, PCA, EOFs

Climate Modelling: CMIP5/CMIP6, bias correction, downscaling, RCP & SSP scenarios; Scope 1/2/3 accounting

Geospatial: xarray, geopandas, rasterio, GDAL, netCDF/GRIB/HDF5; satellite imagery (MODIS, GOES, Landsat, Copernicus); Google Earth Engine

Risk & Disclosure: Physical, transition and supply-chain risk; earnings-at-risk; double materiality; TPT, CSRD, CSDDD, TCFD/ISSB, TNFD, SECR, GRI, CDP

Leadership: Team Lead managing modellers, scientists and engineers; product strategy; trusted advisor to C-suite, boards and government stakeholders across Bloomberg, Cambridge, GFANZ, FTSE100 and the UN

WORK EXPERIENCE

Lead Environmental Risk Modeller

Risilience • London, UK • Mar 2025 to Present

- **Team Lead** for climate and nature risk across some of the world's biggest food, fashion, and entertainment brands, alongside global asset managers, underwriters and FTSE100 clients.
- Lead the planning and build for a full-company digital twin that assesses financial and climate risk due to drought, freeze, heatwave, water stress, agricultural yield and nature depletion.
- **Product Lead** for supply-chain and agricultural risk models spanning 65 agricultural crops, fed by CMIP6 ensembles with bias correction, satellite data and academic sources.
- Designing and delivering a **first-of-a-kind water balance model** for use by clients including Nestlé, Mondelez, Primark, Inditex and Barry Callebaut to quantify water stress across global supply chains.
- Lead the company's **applied AI initiatives** in agricultural risk: built agentic workflows with **LangSmith and LangGraph** that cut model release cycles from 8 weeks to 4; trialling **Gemini Enterprise** (Vertex AI, NotebookLM, BigQuery) for internal research and client-facing analytics.
- Write production code that powers digital twins, what-if analyses and bespoke client engagements; partner with Head of Product on roadmap, releases and client feedback.
- Manage a cross-functional team of modellers, economists, data scientists and engineers.

Senior Climate Risk Modeller

Sillion • London, UK • Dec 2023 to Feb 2025

- Led development of climate risk models for FTSE100 / FTSE250 clients, integrating results into TPT and CSRD reporting and into annual reports across SECR, TCFD/ISSB, CDP and GRI frameworks.
- Built systems to quantify **earnings-at-risk** under multiple scenarios; provided scenario analyses, transition risk assessments and double materiality insights.
- Built the firm's **first transition-planning dashboard** combining economic and climate models, supporting client reporting under TPT, CSRD and CSDDD.
- **Product Lead** for an ESG research tool tracking KPIs across 600+ companies; integrated outputs into Power BI, Tableau and Excel workflows.
- Drove business growth and recurring contracts via professional networks and strategic partnerships; presented to C-suite, boards and press; trained in press and media handling.
- Built Python-based risk models that integrated directly into financial dashboards.
- Managed and mentored junior analysts and consultants, aligning their work with strategic goals in climate risk and sustainability.
- Featured by the British Science Association as part of their Smashing Stereotypes 2025 campaign for my work in STEM and climate.
- Featured by St Catharine's College, Cambridge in their Net Zero by 2040 alumni series (Jan 2025).

Net-Zero Data Lead

Bloomberg L.P. (secondment with GFANZ & CDU) • London, UK • Jun 2022 to Nov 2023

- Led data modelling and engineering for the Climate Data Utility (formerly the NZDPU), a free climate-transition data platform.
- Worked for GFANZ to build a climate report model with key stakeholders including Mark Carney (UN Special Envoy at the time).
- Worked with President Emmanuel Macron, the governments of Switzerland and Singapore, CDP, UNFCCC, and other global organisations.
- Managed a team of 40 analysts for GFANZ's data launch at COP27.
- Built ETL pipelines and APIs ingesting data across hundreds of providers; wrote transformation scripts and data acquisition workflows.
- Co-authored two reports for GFANZ and NZDPU with partners from ISSB and IEA.
- Helped secure the CDP partnership bringing 10,000+ companies onto the platform.
- Supported the CTO on RFP processes, vendor evaluation and management of engineering contractors; helped author the NZDPU RFP.
- Delivered briefs on ~700 climate KPIs used in Michael Bloomberg's keynote address for COP27.
- Subject-matter expert on Scope 1, 2 and 3 GHG emission models, global reporting standards and jurisdictional policies.

Climate Science Researcher

University of Cambridge & British Antarctic Survey • Cambridge, UK • Oct 2021 to Jun 2022

- Fully-funded research at the Scott Polar Research Institute on atmospheric drivers of Antarctic surface melt.
- Analysed ~1.7 billion geospatial data points across CMIP5 GCMs, RCMs and in-situ observations from NOAA and Copernicus (ESA).
- Developed a Convolutional Neural Network and ran EOF analysis to identify seasonal melt patterns; published findings and presented at conferences.
- Tech stack: Python, xarray, pandas/geopandas, scipy, pyproj, netCDF, HDF5, rasterio, GDAL, Shapely, sklearn, EOF, fiona, plotly, cartopy, matplotlib.
- Featured on Cambridge's official channels; my work is used as course material in MPhil-level teaching.
- Supervised by Dr Amelie Kirchgaessner (British Antarctic Survey) and Professor Ian Willis (Scott Polar Research Institute).

Brand Manager, Sustainability

P.G.A. Hospitality • Vancouver, Canada • Sep 2018 to Oct 2021

- Worked as an independent contractor with the CEO to drive sustainable business development; led a team of 10.
- Achieved significant growth: increased revenue by 233%, raised CAD 1.5M in investment, increased profits by 31%.
- Proactively secured contracts with local, sustainable suppliers to mitigate supply chain challenges during the pandemic.
- Developed business plans, marketing materials and seasonal menus.
- Wrote and won an RFP bid for a new location.
- Earned 9 awards and a position in Greater Vancouver's Top 30 cafes.

Lead Teaching Assistant, Computer Science

University of British Columbia • Vancouver, Canada • Dec 2016 to Jun 2020

- Managed a team of 18 TAs as Head Teaching Assistant of Computer Science.
- Ran labs and lectures for ~1,000 students in Python and C++; created and graded course modules and labs; awarded Best TA three years running.
- Helped pilot a new data science course; authored modules on Climate Data Computing and contrast stretching using open-source satellite imagery.

Research Assistant, Atmospheric Science

University of British Columbia • Vancouver, Canada • Dec 2016 to May 2020

- Worked across multiple research labs in the Earth, Ocean and Atmospheric Sciences department.
- Studied glacial katabatic winds and cloud-street formation over Greenland using MODIS, GOES-16/17 and Landsat 7/8 imagery and ECMWF reanalysis data.
- Built ETL pipelines on AWS EC2/S3; ran GOES data via Jupyter notebooks on EC2 instances pulling from NOAA's S3 bucket.
- Applied PCA, Sobel edge detection, Self-Organizing Maps and CNNs to mesoscale convection problems and melt-season streamflow detection across Canada.
- Conducted eddy-covariance measurements during the Buckley Bay wildfires; tested diffused-radiation fertilization against NASA AOD data (contributed to McKendry et al., 2019).
- Collected, identified and analysed phytoplankton in the Burrard Inlet; studied distribution, chlorophyll levels and carbon-sink role.
- Analysed plankton distributions, temperature and salinity using MATLAB to investigate climate change's impact on productivity.
- Constructed and set up weather sensors to study blue carbon and the urban heat island effect; trained on calibrating and maintaining meteorological field instruments to WMO standards.

Geospatial Engineer

UAViation Aerial Solutions • Vancouver, Canada • Jul 2018 to Sep 2019

- Team Lead for the software team responsible for flight-automation app development.
- Project Lead on LiDAR drone mapping operations and GIS post-processing across BC, Alberta and Yukon.
- CEO's right hand in client meetings; my work contributed to a successful exit (acquired by Volatus Aerospace).
- Worked with Android Studio, REST APIs, Google Maps API and DJI's UX SDK; used Pix4D, ArcGIS and QGIS for geospatial post-processing.
- Used C++, Java and Python to tackle photogrammetry and LiDAR flight-mapping problems.
- Delivered 18 projects in 12 months for government, search-and-rescue, BC Wildfire Services, natural resource sectors, academia and private industry.

EDUCATION

MPhil in Polar Studies (Atmospheric Physics)

University of Cambridge, St Catharine's College • Cambridge, UK • 2021 to 2022

- **Scott Polar Scholarship:** full scholarship awarded to the top applicant.
- Dissertation: investigating atmospheric drivers of surface melt on the George VI Ice Shelf, Antarctica, over 32 years; analysed up to 1,734,576,480 geospatial data points; built a CNN and EOF analysis to predict clustered weather patterns influencing melt.

BSc in Atmospheric Science

University of British Columbia • Vancouver, Canada • 2015 to 2020

- Graduated top of class.
- **Dean's Honour Roll** (2017, 2018, 2019, 2020).
- **Ministry of Education and BC Excellence Scholarships** (top 10% of students starting post-secondary education in British Columbia).
- Undergraduate thesis: significance of glacial katabatic winds on cloud formation over Greenland, using MODIS, Landsat 7/8 and GOES-16/17 imagery, image clustering, PCA, Sobel edge detection, neural networks and Self-Organizing Maps. Supervised by Dr Ian McKendry. Thesis score: 93%.

BOARD APPOINTMENTS

Board Member, Revenue Generation

Compassionate Eye Foundation • Remote • Aug 2024 to Feb 2026

- Serve on the board of a global charity to develop and deliver a revenue generation strategy aligned with CEF's long-term vision.
- Help build strategy to deliver projects supporting the UN Sustainable Development Goals and improve standard and cost of living at a grassroots level.
- Research, identify and apply for grants, trusts and major funding opportunities.
- Build and maintain relationships with key donors, corporate sponsors and stakeholders.
- Explore new revenue streams and create roadmaps; work closely with communications and project teams to support funding applications.

Co-Organiser & DEI Lead

Polar Impact • Cambridge, UK • Oct 2021 to Mar 2023

- Co-ran an NGO supporting BIPOC professionals in the polar research community.
- Helped write grant proposals that won a cumulative £50,000 from the National Science Foundation, UKRI and National Geographic.
- Organised COP26 climate-justice speaker panels.
- Featured in St Catharine's College's alumni publications for this work.

SELECTED FEATURES & PUBLICATIONS

- Smashing Stereotypes 2025 — British Science Association profile (Mar 2025).
- Tackling Eco-Anxiety — invited guest blog for the British Science Association, Mental Health Awareness Month (May 2025).
- Net Zero by 2040 Alumni Series — St Catharine's College, University of Cambridge (Jan 2025).
- Amplifying Diversity in Polar Exploration — St Catharine's College, University of Cambridge (Nov 2021).
- Co-author, GFANZ Recommendations and Guidance on Financial Institution Net-Zero Transition Plans (Nov 2022).

- Co-author, Development of the Net-Zero Data Public Utility (Nov 2022).

AWARDS

- **2021:** Scott Polar Scholarship — full scholarship to study at the University of Cambridge's Scott Polar Research Institute.
- **2017–2020:** Dean's Honour Roll — awarded for scoring above 85% (four consecutive years).
- **2018–2020:** Best Teaching Assistant Award (three consecutive years).
- **2016:** Ministry of Education Scholarship — top 25% of students starting post-secondary education in Canada.
- **2016:** BC Excellence Scholarship — top 10% of students starting post-secondary education in British Columbia.
- **2015:** Bronze Medal, Canadian National Science Fair.
- **2015:** Western University Entrance Scholarship.
- **2015:** Chief Judge Award, Vancouver Provincial Science Fair.
- **2015:** Gold Medal, Vancouver Provincial Science Fair.
- **2015:** UBC Engineering Physics Award, Vancouver Provincial Science Fair.
- **2014:** Top 25, Pascal Mathematics Contest.

SELECTED RESEARCH PROJECTS

MPhil Dissertation, University of Cambridge

Investigating atmospheric drivers of surface melt on the George VI Ice Shelf (GVIIS) in Antarctica over 32 years. Analysed geospatial datasets of up to 1,734,576,480 points; worked with large multi-dimensional data (netCDF, GRIB, TIFF) and Python libraries for mapping and visualisation (rasterio, iris, pandas, scipy, sklearn, xarray, cartopy, matplotlib, pyproj, fiona, shapely, geopandas, GDAL). Built a Convolutional Neural Network and conducted EOF analysis to predict clustered weather patterns influencing melt on the GVIIS. Supervised by Professor Ian Willis (Scott Polar Research Institute) and Dr Amelie Kirchgaessner (British Antarctic Survey).

Undergraduate Thesis, University of British Columbia

Studied the significance of glacial katabatic winds on cloud formation over Greenland. Used satellite imagery from MODIS, Landsat 7/8 and GOES-16/17, image clustering, Principal Component Analysis (PCA), Sobel edge detection, a custom neural network, and Self-Organizing Maps to detect mesoscale cellular convection. Supervised by Dr Ian McKendry. Final score: 93%.

Plant Productivity During the Buckley Bay Wildfires

Modelled plant productivity during the Buckley Bay wildfires using NASA Aerosol Optical Depth data to test diffused-radiation fertilisation (contributed to McKendry et al., 2019). Used to study the carbon cycle during wildfire events across British Columbia.

AWS-Hosted GOES Cloud Computing Pipeline

Built ETL pipelines on AWS EC2 servers and virtual platforms to ingest GOES data via Jupyter notebooks pulling from NOAA's S3 bucket. Produced true and pseudo-colour imagery, climate animations, and maps of rain rate, radiances and reflection for deep convective storms.

Carbon Flux Weather Stations, Vancouver

Constructed and set up weather stations to detect carbon flux from bogs in Vancouver using eddy covariance methods. Trained on operating, calibrating and maintaining meteorological field instruments to WMO standards.

Hybrid Photovoltaic-Thermoelectric Solar Cell

Invented a hybrid photovoltaic and thermoelectric solar cell that produced electricity at 19% higher efficiency than conventional cells. Excess heat from the photovoltaic cell was absorbed by the heating side of a Peltier

tile, while the cooling side was passively cooled by a thermosiphon to reduce hot spots. Selected as the top project in the Innovation category from over 900 projects nationwide at the Canadian National Science and Engineering Fair.

CERTIFICATIONS

- Data Engineer in Python — DataCamp, Dec 2024.
- Software Engineering Principles in Python — DataCamp, Dec 2024.
- Foundations of Git — DataCamp, Dec 2024.
- Streamlined Data Ingestion with pandas — DataCamp, Dec 2024.
- Understanding Cloud Computing — DataCamp, Dec 2024.
- Data Design — DataCamp, Dec 2024.
- Intermediate Importing Data in Python — DataCamp, Dec 2024.
- Intermediate SQL — DataCamp, Dec 2024.
- Introduction to Relational Databases in SQL — DataCamp, Dec 2024.
- Bloomberg Market Concepts — Bloomberg, Nov 2023.
- Python Data Science Toolbox (Part 2) — DataCamp, Aug 2023.
- Python Data Science Toolbox (Part 1) — DataCamp, Jun 2023.
- Corporate Finance Foundations — LinkedIn, Dec 2022.
- Intermediate Python — DataCamp, Dec 2022.
- Intermediate SQL — DataCamp, Dec 2022.
- **Carbon Literacy Certificate** — The Carbon Literacy Project, Dec 2021.